MAJOR PROJECT · SESSION 2026 - 2027

STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

A Project Report Submitted to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal
towards the partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering

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Live demo
Scan, or open
aicad-vercel.vercel.app

PART 1

The problem

Designing a standard joint is a fixed sequence of formulas. The sequence is the slow part, and it repeats in full every time the load changes.

WHERE THE TIME GOES

Seven steps by hand, and every one repeats when the load changes

STEP PERFORMED BY HAND	REPEATS
Rod diameter from direct tension	yes
Pin from double shear	yes
Pin from bending, adopt the larger	yes
Seven proportions from the rod diameter	yes
Nine failure checks against allowables	yes
Re-size whatever fails, check again	yes
Four views, dimensions, title block	yes

Parametric CAD does not solve this. A modeller regenerates whatever dimensions it is given. It will not object if the pin is too slender to carry the bending moment placed on it.

The engineering is the small part. Choosing allowable stresses and interpreting the result takes minutes. Evaluating the sequence takes the afternoon.

Objective. Keep every formula exactly as the textbook has it. Remove only the repetition.

PART 2

Literature review

Three published sources define the design procedure used here. Two of them are out of copyright, so every formula can be shown to a reviewer.

LITERATURE REVIEWED

What each source contributes, and why three were needed

SOURCE	YEAR	WHAT IT PROVIDES	USED FOR
NPTEL, IIT Kharagpur Design of Machine Elements I, Module 4	Course	Complete design procedure for the socket and spigot cotter joint and the knuckle joint: empirical proportions as multiples of the rod diameter, and the full list of failure modes. Lesson 1 gives the key equations, Lesson 4 the bolt tightening relation.	Primary procedure for four of the six joints
Unwin *The Elements of Machine Design*	1899	Page 179: division of the slot breadth between cotter and gib, thickness as a fraction of that breadth, and the permissible taper range.	The gib and cotter joint, which the course does not cover
Leutwiler *Elements of Machine Design*	1917	Article 98 lists ten distinct ways a cotter joint fails, with the expression for each. Article 275 analyses the flange coupling as key, bolts and flange at the hub.	Independent second opinion, and the coupling

Why a second source on the same joint matters. Where two independent texts published eighteen years apart agree, the expression can be treated as settled. Where they disagree, the disagreement is itself a finding, and three such findings are reported later in this deck.

Standards consulted. SP 46 for first angle projection and the title block, IS 2048 for parallel keys and keyways, IS 1363 and IS 1364 for hexagon head bolts and nuts, IS 2693 for cast iron flange couplings.

EXISTING APPROACHES, AND THE GAP

Modelling is solved. Language is solved. The connection between them is not.

Parametric CAD

A model driven by a parameter table regenerates when the parameters change, which removes the drafting labour.

It does not determine the parameters. The designer still computes them, and the software regenerates an unsafe model as readily as a safe one.

Language models writing CAD code

Recent work generates modelling code directly from a description. Geometry appears quickly.

No dimension can be traced to a calculation and no failure mode is verified. Geometry that cannot be justified is of no use in a design review.

Expert systems and handbooks in software

Reliable within the range they were written for, and long established in industry.

The calculation is hidden. A result cannot be checked against a published source without access to the implementation.

The gap addressed here. A system in which a written requirement is interpreted by a language model, the resulting parameters drive a deterministic calculation taken from published literature, and that calculation stays open to inspection by the reader.

PART 3

What was built

A written requirement goes in. A sized, checked, modelled and drawn joint comes out, with the whole working printed.

THE SYSTEM

Five stages, each reporting its result to the next

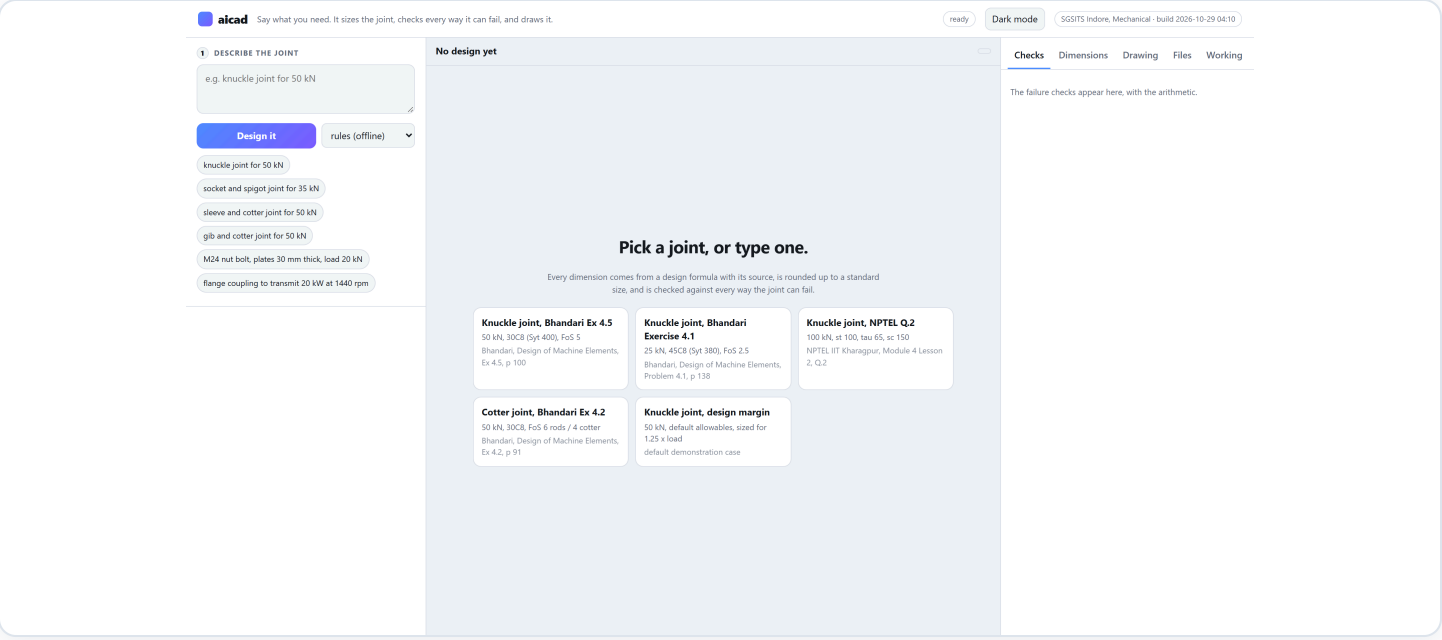
01 · AI
Read
Sentence to parameters: joint, load, allowable stresses.

02
Size
Governing formula, rounded up to a preferred size.

03
Check
Every failure mode against its allowable, iterate on failure.

04
Model
Exact solids, STEP files, assembly in position.

05
Draw
First angle A3 sheet, dimensions, title block, DXF.



The entry screen. Design rules are held as data, not code, so the source of any dimension can be reported the moment it is computed.

6
JOINTS IMPLEMENTED

44
DISTINCT FAILURE CHECKS

301
AUTOMATED TESTS

THE DECISION THAT MATTERS

The AI reads the requirement. It selects no dimension and performs no check.

Reading `50 kN` as 50,000 N, and *socket and spigot* as the cotter joint, is a language problem. A language model is good at that.

Choosing a pin diameter is an engineering problem with one correct answer from a formula. That stays with deterministic code traceable to a printed page.

Safeguard. Any number the model returns that does not appear in the request is discarded and the user is told, so an invented figure cannot enter the calculation.

Why not generate the whole model with AI? Because then no dimension could be checked against anything. Geometry that cannot be justified is of no use in a design review.

STAGE	PERFORMED BY	CHECKABLE AGAINST
Read the requirement	**Language model**	the sentence
Size every dimension	formula	cited book page
Verify failure modes	formula	cited book page
Build the solids	CAD kernel	computed dimensions
Draw the sheet	FreeCAD	assembled geometry

SCOPE

Six joints, 44 failure checks, every one satisfied

JOINT	PARTS	DIMENSIONS	CHECKS	DUTY	BUSIEST UTILISATION OF ALLOWABLE
Knuckle joint	5	13	9	Load P = 50,000 N	79%
Socket and spigot cotter joint	3	14	10	Load P = 35,000 N	76%
Sleeve and cotter joint	5	12	8	Load P = 50,000 N	77%
Gib and cotter joint	4	20	8	Load P = 50,000 N	79%
Hex bolt, nut and washer	3	11	2	Load P = 20,000 N	71%
Protected type rigid flange coupling	12	15	7	Torque T = 500,000 N mm (500 N m)	79%

Busiest check sits at 67 to 79 percent. That band is the design factor working: each joint is sized for 1.25 times the load, then reported at the load, so margin is left in hand rather than spent on paper.

The gib and cotter joint needed a decision. Unwin gives the breadth split and thickness but no strap proportions. Rather than borrow or invent a ratio, every dimension the source omits is sized against a failure check.

PART 4

Worked example

A knuckle joint for 50 kN, shown exactly as the software computes it.

WORKED EXAMPLE · 1 OF 3

Which formula governs, and why the larger requirement wins

COMPUTED BEFORE ROUNDING	MM	EFFECT
Rod in direct tension	32.57	rod adopted as **35**
Pin in double shear	25.75	not governing
Pin in bending	44.85	governs, pin adopted as **45**

$D = \sqrt{(4P / \pi\sigma_t)}$ $d = \sqrt{(2P / \pi\tau)}$ $d = [16P(t_1/3 + t/4) / \pi\sigma_b]^{1/3}$

Bending asks for 1.7 times what shear asks for. A pin sized from shear alone passes its own check and bends in service. The source takes the same step.

SYMBOL	MEANING	MM	RULE AND SOURCE
D	rod diameter	35	rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred size
d	pin diameter	45	governed by pin bending
d2	eye outer diameter	90	d2 = 2.0 x d, the pin diameter (outer diameter of eye and fork, d0 = 2d), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq 4.26j]
d3	pin head and collar diameter	70	d3 = 1.5 x d, the pin diameter (pin head and collar diameter, 1.5d), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq 4.26o]
D1	enlarged end, 1.1D	40	D1 = 1.1 x D, the rod diameter (enlarged rod diameter near the eye and fork, D1 = 1.1D), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq 4.26b]
t	eye thickness	45	t = 1.25 x D, the rod diameter (thickness of eye of rod B, b = 1.25D), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq 4.26n]
t1	fork plate thickness	30	t1 = 0.75 x D, the rod diameter (thickness of each fork eye, a = 0.75D), rounded up to a standard size [Bhandari, Design of Machine Elements, sec 4.12, eq 4.26m]
t2	pin head thickness	18	t2 = 0.5 x D, the rod diameter (thickness of pin head), rounded up to a whole mm [NPTEL IIT KGP M4L2 fig 4.2.3.1]

WORKED EXAMPLE · 2 OF 3

All nine failure modes, evaluated at the actual load

FAILURE MODE	EXPRESSION	STRESS MPA	ALLOWABLE	USED	UTILISATION
rod tension	`4P / (pi D^2)`	52	75	69%	
pin double shear	`P / (2 pi/4 d^2)`	15.7	60	26%	
pin bending	`32M / (pi d^3), M = P/2 (t1/3 + t/4)`	59.4	75	79%	
eye tension	`P / ((d2 - d) t)`	24.7	75	33%	
eye shear	`P / ((d2 - d) t)`	24.7	60	41%	
eye crushing	`P / (d t)`	24.7	150	16%	
fork tension	`P / ((d2 - d) 2 t1)`	18.5	75	25%	
fork shear	`P / ((d2 - d) 2 t1)`	18.5	60	31%	
fork crushing	`P / (d 2 t1)`	18.5	150	12%	

Nine of nine satisfied. Busiest is shear of the eye at 76 percent of its allowable stress.

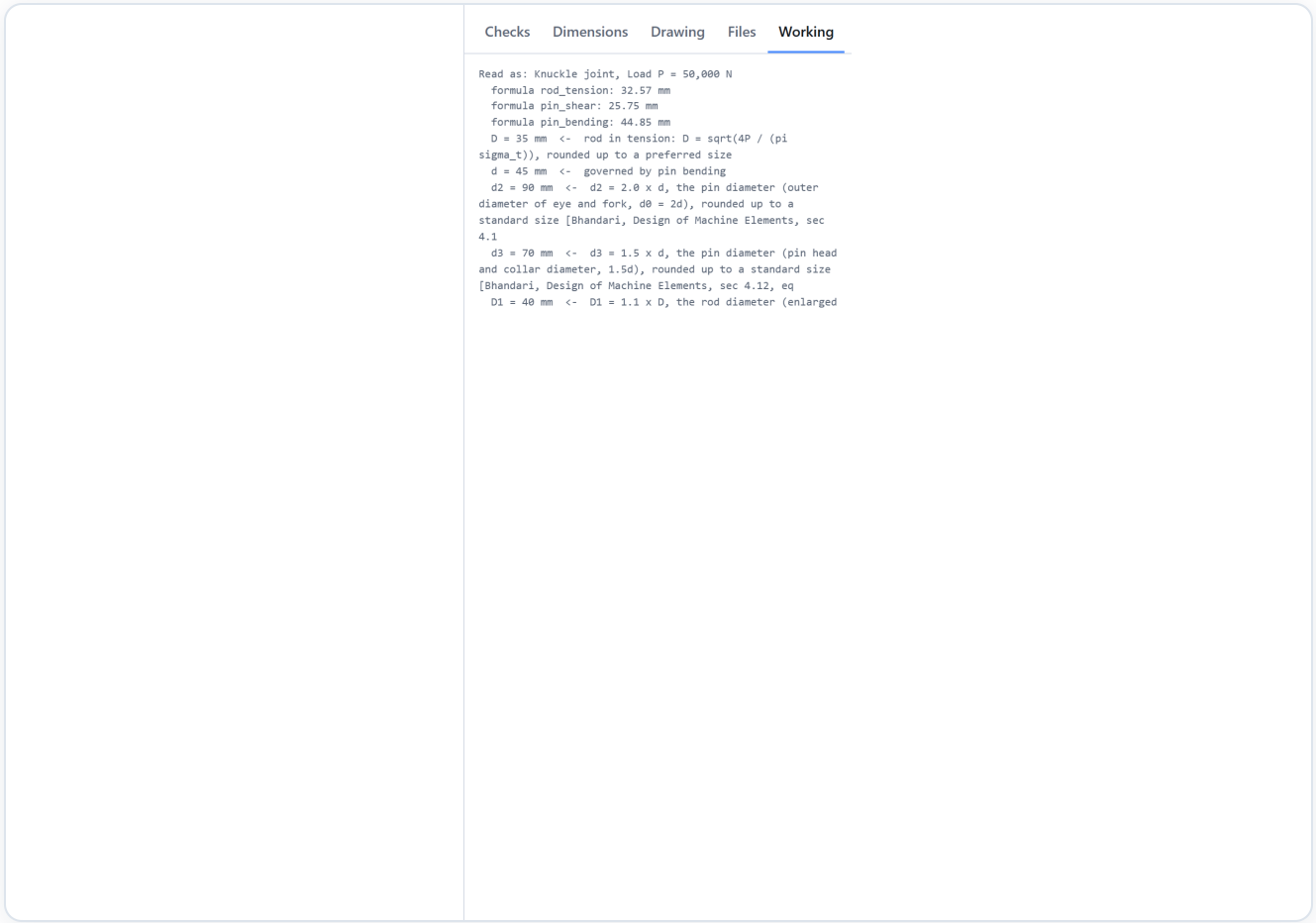
None of this is stored. Ask for 80 kN and every value in the table changes, and the pin is re-sized.

The working is printed, so the output can be checked rather than trusted

Each run prints the values computed before rounding, the size adopted, the rule and source behind every proportion, every check with its arithmetic substituted, and the volume and mass of each solid.

What this buys in a review. A reviewer does not have to accept the result. Every line can be recomputed by hand from the formula printed beside it.

It is also the reason the next three slides exist. Printing every check is what made the errors in the published sources visible.



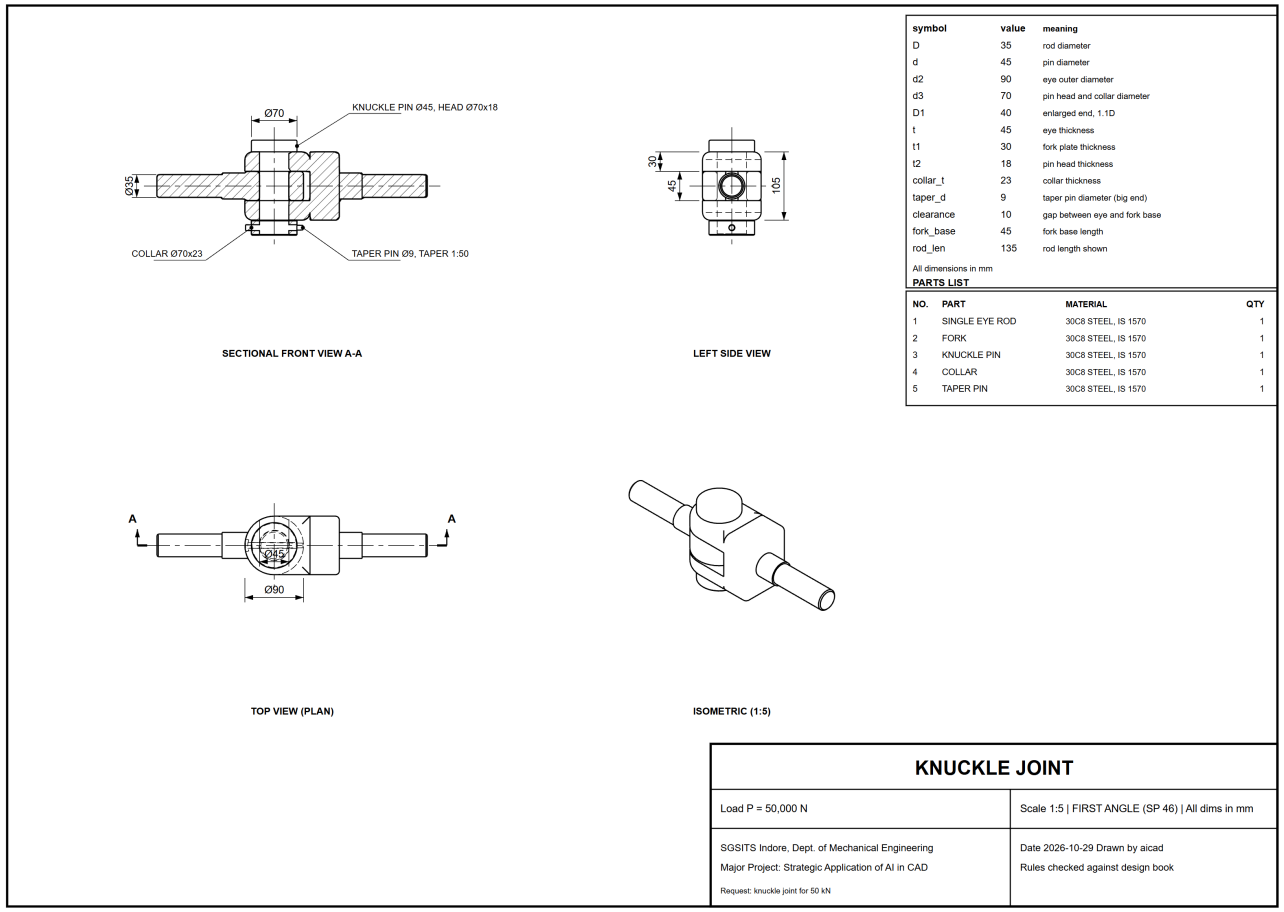
Working printed for the example above.

PART 5

Output

Exact solids in a neutral format, and a first angle sheet produced without opening a CAD application.

First angle A3 sheet, dimensioned, with a title block



How it is made. FreeCAD is driven without its graphical interface. The assembled geometry is projected to obtain the front view, the left side view and the plan, arranged in first angle projection, with an isometric view.

FILE	USE
`drawing.svg`	A3 sheet
`drawing.dxf`	AutoCAD
`assembly.step`	NX, SolidWorks, Fusion
`parts/*.step`	each component
`*.FCStd`	editable FreeCAD file

OUTPUT · NX10

Verified the file structure. Opening it in NX10 itself is the one step still in the lab.

What we checked, directly, not assumed. The exported assembly was opened as text and read. Two things decide whether an assembly import works at all, and both are confirmed below.

```
FILE_SCHEMA(('AUTOMOTIVE_DESIGN { ... 214 ... }'));

#41  = PRODUCT('rod','rod', ...);
#784 = PRODUCT('strap','strap', ...);
#2141 = PRODUCT('cotter','cotter', ...);
#2490 = PRODUCT('gib','gib', ...);
```

Exact lines read from assembly.step for the gib and cotter joint.

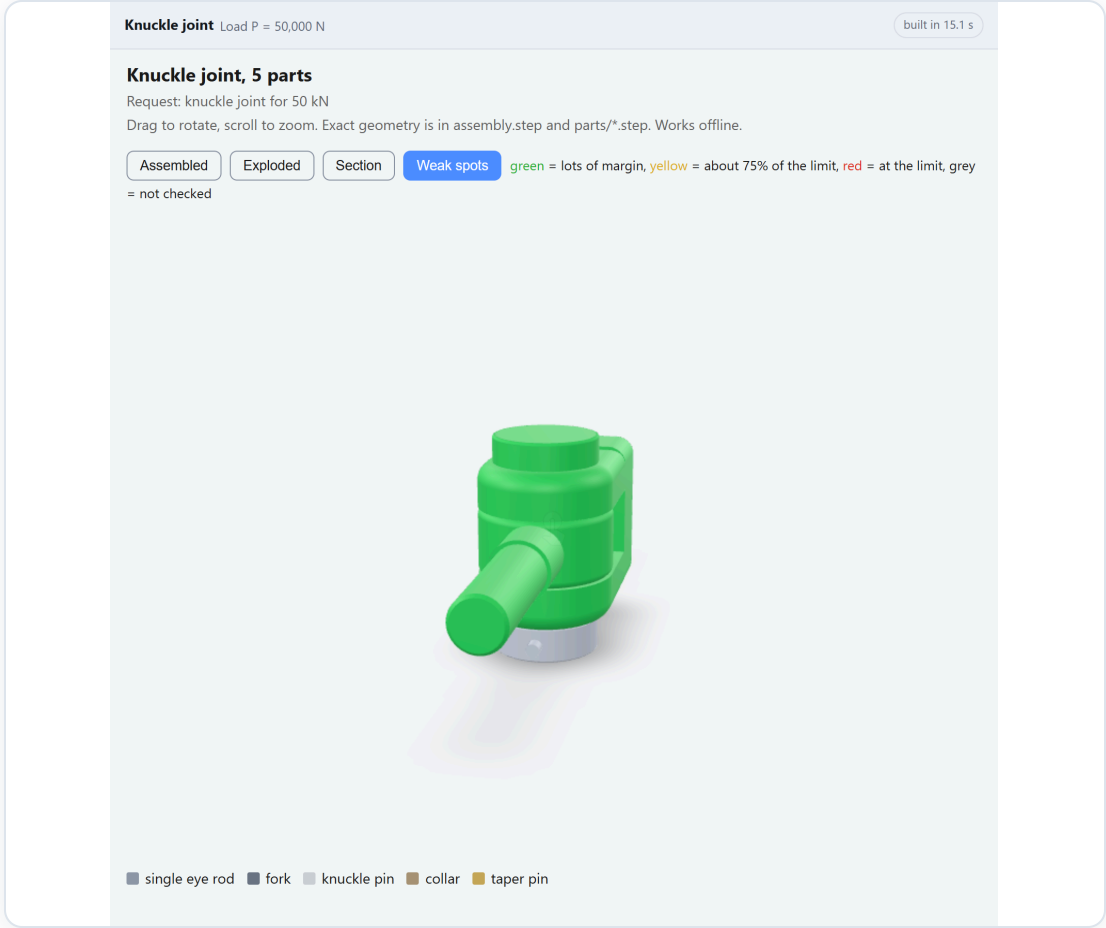
CHECK	RESULT
Schema is AP214 (AUTOMOTIVE_DESIGN)	confirmed, exactly what NX10 reads
Parts stay separate, not fused	confirmed, 4 named solids for 4 parts
Opens in NX10 as an assembly	not yet observed, lab-only software

Said plainly, not glossed over. This is the one objective of six, from Section 1.3 of the report, not yet confirmed by watching it happen. The file is built to the structure NX10 needs; nobody has opened it there yet.

Assembled, exploded, and coloured by how close each part is to its limit



Assembled model, rotatable in the browser, exact geometry in the STEP files.



The same joint coloured by utilisation, so the member nearest its limit is visible without reading the table.

PART 6

Verification

Every proportion and every check compared against published design literature. The comparison found three errors in the sources.

SOURCES

Checked against three published sources, two of them out of copyright

JOINT	SOURCE CONSULTED	OUTCOME
Knuckle joint	Bhandari Ex 4.5; NPTEL M4 L2	Ex 4.5 reproduced exactly: 7 of 7 sizes, 6 of 6 stresses
Socket and spigot cotter	Bhandari Ex 4.2; NPTEL M4 L2; Leutwiler Art. 98	Ex 4.2: 9 of 10 sizes; ten checks verified against two books
Gib and cotter	Unwin (1899), page 179	breadth split, thickness and taper verified
Flange coupling	NPTEL M4 L1 keys; Leutwiler Art. 275 bolts	all seven checks verified twice; proportions not verifiable
Sleeve and cotter	four books searched, none covers it	checks verified by correspondence; proportions unverified
Hex bolted joint	NPTEL Module 4 Lesson 4	tightening relation verified; size table unverified

Where no source exists, the software says so on every run. Leutwiler records that no standard set of flange coupling proportions was ever agreed, and that published tables are individual manufacturers' practice. Those proportions are marked unverified rather than given a source they do not have.

FINDING 1 OF 3

A worked example that fails its own check

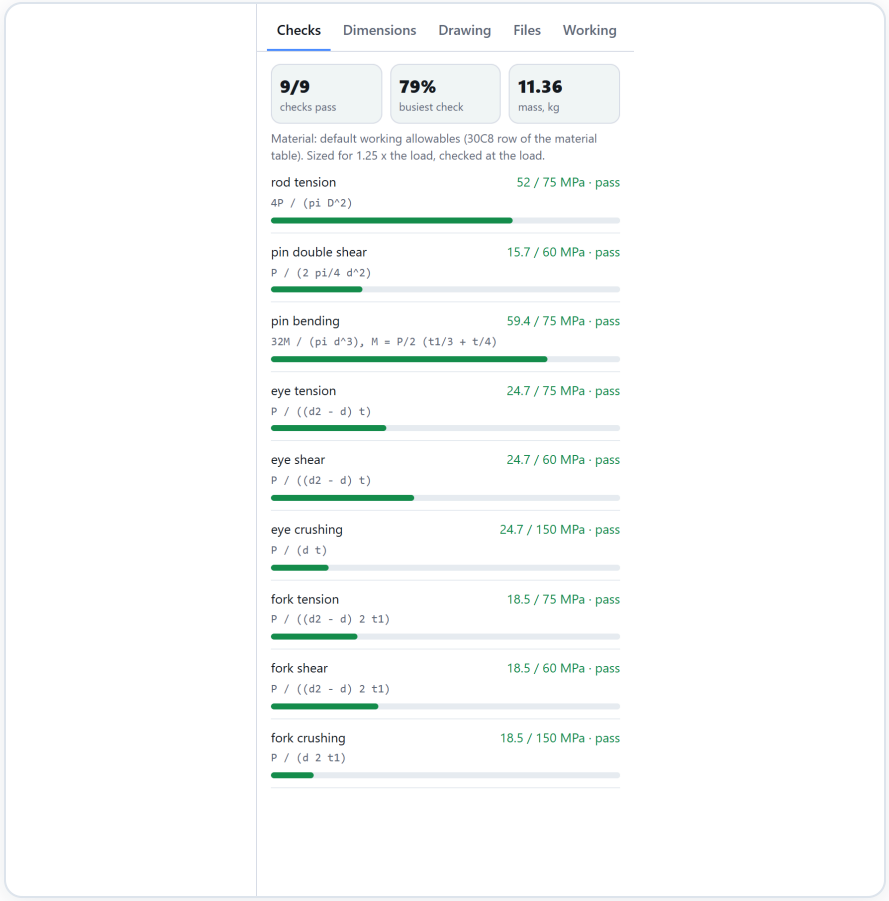
The knuckle joint worked problem raises the pin from **40 mm to 55 mm** for bending, then evaluates the eye using the **original 40 mm pin**.

```
eye shear = P / ((d₂ − d) t)

with d = 55   100000 / ((80 − 55) × 50) = 80 MPa
allowable   65 MPa
```

> **Its own expression, with the pin it actually adopted, exceeds its own allowable by 23 percent.**

What this software does instead: it enlarges the outer diameter of the eye until the check passes, and reports that it has done so.



Every check shows its substituted arithmetic, which is how an inconsistency of this kind becomes visible rather than inherited.

FINDINGS 2 AND 3 OF 3

An omitted failure mode, and a constant printed wrongly

Omitted failure mode

The cotter joint procedure verifies crushing of the socket but not crushing between the cotter and the spigot.

$$P / (d_1\, t) = 50000 / (30 \times 10) = \mathbf{167\ MPa}$$
$$\text{allowable} \quad 110\ \text{MPa}$$

Leutwiler, Article 98, lists ten failure modes and includes this one. Two independent texts, eighteen years apart, contain a check the lesson omits. That is what makes it an omission and not a preference.

Constant printed wrongly

The initial tightening load is printed as $284d$ kN. The worked example on the same page obtains 56.8 kN for an M20 bolt.

$$284 \times 20 = 5{,}680\ \text{kN} \neq 56.8\ \text{kN}$$
$$2840 \times 20 = 56{,}800\ \text{N} = 56.8\ \text{kN}\ \checkmark$$

The page refutes itself. The constant is $2840d$ newtons and the printed relation is out by a factor of one hundred. Every bolted design now reports the tightening load.

PART 7

Validation and limits

An independent finite element cross check, the tests that hold the work together, and what the system does not do.

INDEPENDENT CROSS CHECK

Finite element analysis, and knowing when not to believe it

CASE	FORMULA	COMPUTED	DIFFERENCE
20 mm bar, 10 kN tension	31.83	32.22	+1.2 %
Knuckle eye rod, 50 kN	52.0	57.9	+11.3 %

The bar validates the method. The eye rod reads higher because the formula treats it as a plain bar and ignores the stress raised at the pin bore. Both stay under the 75 MPa allowable.

A peak that never settles is not a stress. On one component the computed peak moved between **157 and 191 MPa** across a sixteen fold increase in nodes. That is a singularity at a constrained face, not a value to design to.

So the software re-solves on a mesh of half the element size whenever the peak exceeds 1.5 times the calculated value, and reports `NOT CONVERGED` if it then moves more than 5 percent.

Stated plainly in the report. One linear static run on one component with boundary conditions the software chooses. A second opinion on the formula, not a replacement for analysis.

CORRECTNESS

301 automated tests, and the row that matters most is the last one

WHAT IS TESTED	WHY
Sizing arithmetic	values match the worked problems in the sources
Generated geometry	each part is one valid solid, no two parts overlap
Request interpretation	parameters correct, invented numbers dropped
Drawing sheet	sheet, DXF and FreeCAD file are produced
Finite element path	face selection refers to dimensions that exist
The three findings	**assert the source errors are real**

Why that last row exists. Tests assert that the published answers fail their own checks. Without them, someone could later correct the software back to the printed values and every other test would still pass.

A test that found what no dimension check could. Comparing solids pairwise caught a gib hook modelled 1.25 mm inside the strap, occupying metal another part already occupied. Every individual dimension was correct.

LIMITS

What this does not do, said before anyone asks

- **Static loading only.** Fatigue, endurance limit and stress concentration factors are outside the scope.
- **Nominal dimensions only.** Tolerances and fits are not applied, so the sheet is not yet a manufacturing drawing.
- **Three of six joints have unverified proportions.** The sleeve and cotter joint, the flange coupling and the hex bolt size table. The software reports this on every run.
- **Nut thread stripping is not checked** for the bolted joint.

Scope for future work

- Riveted and welded joints, which follow the same pattern.
- Fatigue loading, through an endurance limit and a stress concentration factor.
- Tolerances and fits, to make the sheet usable for manufacture.
- A journal script for Siemens NX, so the assembly is built inside it rather than imported.

OBJECTIVES REVISITED

Six objectives set at the start, assessed against the evidence, not restated as met

#	OBJECTIVE	STATUS	EVIDENCE
1	Interpret a plain English requirement	**Met**	Language model extracts joint and load; anything not traceable to the sentence is dropped.
2	Size and verify every failure mode, with iteration	**Met**	All 44 applicable checks satisfied; iterative enlargement closes gaps the source proportions leave.
3	Exact solids in a neutral format, opening in commercial CAD	**Substantially met**	AP214 schema and 4 separate named solids confirmed by direct inspection; opening inside NX10 itself not yet observed.
4	A first angle drawing sheet to Indian practice	**Met for projection and layout**	SP 46 projection and title block done; tolerances and fits not implemented.
5	Verify against literature, record disagreements	**Met for 4 of 6 joints**	2 joints carry proportions with no supporting source, reported on every run, not hidden.
6	Browser use without installation	**Met**	Deployed and reachable without install; hosted copy runs the rule-based reader, not the language model.

Objectives 1, 2 and 6 are met; 3, 4 and 5 are met in part. Each open point is stated in the table: the NX10 import not yet observed, no tolerances on the sheet, two joints partly verified. Reporting those points is itself part of Objective 5: disagreements and open points recorded explicitly, not smoothed over.

TIMELINE

Where the project stands across both semesters

COMPLETE

Phase 1

Semester 7

- Problem formulation and scope
- Literature review: NPTEL, Unwin 1899, Leutwiler 1917
- Design procedure and failure modes extracted

COMPLETE

Phase 2

Semester 7

- Sizing engine and preferred size rounding
- All six joints, 44 failure checks
- Iterative correction when a check fails

CURRENT STAGE

Phase 3

Semester 7

- Solid modelling and STEP export
- First angle A3 sheet and DXF
- Verification against the three sources
- Finite element cross check, 296 tests
- Web deployment, report and presentation

PLANNED

Phase 4

Semester 8, proposed scope

- Riveted and welded joints
- Fatigue loading and stress concentration
- Tolerances and fits on the drawing
- Verify the two proportion sets still without a source
- NX journal script for direct assembly
- Bill of materials and cost

Everything shown in green and amber exists today. Six joints, 44 checks, the drawing sheet, the verification against three books and 296 tests are all working and demonstrated in this presentation.

Semester 8 is a proposal, not a commitment. These items come from the limits listed on the previous slide and are to be agreed with the guide.

PHYSICS FIRST

The AI proposes, the physics decides

1 · AI
Read the sentence
"knuckle joint for 50 kN" becomes joint = knuckle, P = 50,000 N. Any number not in the sentence is thrown away.

2 · PHYSICS
Size from the equations
$D = \sqrt{(4P / \pi\sigma_t)}$, pin from shear and from bending, the larger wins.

3 · PHYSICS
Check every failure
Tension, double shear, bending, crushing: nine checks with the arithmetic printed.

4 · PHYSICS
Fail? Enlarge and recheck
The member that fails grows 1 mm at a time until every check passes.

5 · CAD
Build and draw
Exact solids in STEP, the assembly, and a first angle A3 sheet.

A physics informed neural network (PINN) learns a design and is penalised when it breaks an equation. The penalty makes a violation unlikely, not impossible.

Here the equations are not a penalty, they are the answer. No dimension comes from the model, so no output can break a law of mechanics. The model only reads; the physics sizes, checks and corrects.

CONCLUSION

Every dimension it produces can be traced to a formula, and every formula to a page in a book.

The language model reads the requirement. The engineering stays deterministic, checked, and open to inspection. That division is what allows a natural language interface without any part of the design becoming unverifiable.

6	3	301
JOINTS, 44 CHECKS, ALL SATISFIED	ERRORS FOUND IN PUBLISHED SOURCES	AUTOMATED TESTS

Live demo
Scan, or open
aicad-vercel.vercel.app